

Md Mahamodul Islam

Cloud & Software Engineer — AWS Solutions Architect Associate

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Summary

Cloud and software engineer focused on the delivery pipeline: CI/CD, containerisation, infrastructure as code and observability. Day-to-day experience with Docker, Kubernetes, Terraform, GitHub Actions, Prometheus and Grafana on Linux, alongside Python and Bash automation. AWS Certified Solutions Architect Associate, with cloud fundamentals that carry across providers. MSc in Informatics (Information Security), University of Oslo.

Certifications: AWS Certified Solutions Architect – Associate · AWS Certified Cloud Practitioner · Databricks Certified Data Engineer Associate

Skills

DevOps & Operations: Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Prometheus, Grafana, Infrastructure as Code

AWS & Cloud Architecture: AWS S3, Lambda, EC2, SQS, Glue, Athena, Redshift, Cloud Architecture, Scalability, High Availability, Reliability, Cost Optimization

Software Engineering: C#, .NET Core, ASP.NET Core, Python, FastAPI, REST APIs, Microservices, SQL, Bash

Security & Networking: Zero Trust, OAuth2/OIDC, Keycloak, HashiCorp Vault, mTLS, IAM Concepts, RBAC, Network Security, Secrets Management

Data & AI/ML: Databricks, Apache Spark, Snowflake, ETL/ELT, Data Pipelines, Machine Learning, Deep Learning, NLP, RAG Systems, LLM Integration

Professional: Solution Design, Technical Communication, Requirements Analysis, Technical Presentations, Mentoring, Stakeholder Collaboration, Problem Solving

Professional Experience

Full-Stack Engineer (Data)

May 2025 – June 2026

Nordic Digi Solutions, Oslo, Norway

- Design data-intensive backend and full-stack architectures, translating business requirements and use cases into practical technical solutions.
- Build and maintain backend services and APIs supporting analytical and operational workflows, with focus on scalability, reliability, and maintainability.
- Optimize SQL queries, application workflows, and data models to improve latency and overall system performance.
- Collaborate on technical decisions across application and data components, balancing business requirements with performance and operational considerations.
- Improve data ingestion and transformation workflows to provide reliable information for business decision-making.

Teaching Assistant

Aug 2023 – Dec 2024

University of Oslo, Norway

- Explained complex technical concepts through practical guidance, demonstrations, and one-to-one technical discussions.
- Supported students in debugging applications and evaluating alternative technical approaches to software and database problems.
- Automated grading workflows, reducing manual review effort by 40%.

Data Engineer

Aug 2021 – Jun 2022

Luminous Labs, Dhaka, Bangladesh

- Developed backend and data solutions supporting ingestion and processing of millions of records.
- Improved processing throughput by 60% through application and data workflow optimization.
- Optimized database queries and backend operations, reducing query latency by 35% and implemented CI/CD workflows for reliable application delivery, reducing release failures by 40%.
- Worked across software, database, and infrastructure components to improve system reliability and operational efficiency.

Data Analyst

Apr 2021 – Jul 2021

Coding Object Pvt. Ltd., Kolkata, India

- Analyzed operational datasets and translated findings into actionable recommendations, contributing to a 15% reduction in operational costs.
- Developed automated data workflows to improve reporting accuracy and support business decision-making.

Education

MSc in Informatics (Information Security)

2022 – 2025

University of Oslo, Norway

Focus: Secure Systems Development, Cloud and Microservice Security, Networking, Authentication, Authorization, and Zero Trust Architecture

BSc in Software Engineering

2017 – 2021

Daffodil International University, Bangladesh

Selected Projects

Secure Microservices Platform – Master's Thesis



- Designed a secure microservice architecture using Zero Trust principles with emphasis on identity, endpoint protection, and secrets management.
- Implemented centralized authentication and authorization using Keycloak and OAuth2/OIDC together with HashiCorp Vault for secure secrets management.
- Integrated Prometheus and Grafana for observability and used performance testing to evaluate system behavior under different workloads.
- Applied security and architecture principles relevant to reliable, scalable cloud-native application design.
- **Technologies:** .NET Core, Keycloak, HashiCorp Vault, Docker, Kubernetes, Prometheus, Grafana, OAuth2/OIDC.

End-to-End Cloud ETL Pipeline



- Architected an automated cloud-based data pipeline using Python, Apache Airflow, Snowflake, Docker, REST APIs, and AWS S3.
- Used AWS S3 as a scalable cloud landing zone for extracted data before downstream processing and analytics.
- Implemented automated extraction, validation, orchestration, and historical bulk-loading workflows.
- Containerized pipeline components with Docker to improve portability and operational consistency.
- **Technologies:** AWS S3, Python, Apache Airflow, Snowflake, Docker, REST APIs.

Spark E-Commerce Data Lakehouse



- Designed a scalable data lakehouse using Apache Spark and Databricks with Bronze, Silver, and Gold architecture.
- Implemented incremental loading, deduplication, transformation, and distributed processing for customer and order datasets.
- Applied cloud data architecture principles for scalable storage and analytical processing.
- **Technologies:** Apache Spark, Databricks, Python, SQL, ADLS, Delta Lake.

Research and Publications

- **PhishGuard: A CNN Model for Phishing URL Detection** – Applied deep learning and CNN-based techniques to real-time cybersecurity threat detection.
- **Prediction of Compressive Strength of High-Performance Concrete** – Applied machine learning techniques to infrastructure datasets.
- **Sentiment Polarity Analysis of Bangla Food Reviews** – Developed NLP and sequence-modeling approaches for sentiment classification with 87% accuracy.